ELA: English Language Assistant

An AI-Assisted Framework for Authentic English Comprehension Using Recursive Linguistic Annotation

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github.com/welsol21/FYP_LLM

Problem & Motivation

Mainstream language learning platforms present **pre-simplified content**, stripping the grammatical structures and vocabulary range that define authentic English. Learners are shielded from complexity rather than guided through it.

ELA takes the opposite approach: authentic text and speech are annotated **in place — structure intact** — using rule-based NLP and fine-tuned ML models. Each sentence is decomposed into a **Recursive Linguistic Element (RLE)** tree enriched with CEFR level, tense–aspect–mood tags, IPA, and bilingual pedagogical notes.

The result is a fully offline, client-side system that bridges authentic complexity and learner comprehension — without simplification, cloud dependency, or data upload.

The RLE Data Model

```
Sentence
  └── Phrase (NP/VP/PP/ADJP)
        └── Word: translation
              CEFR · POS · IPA · TAM · note
```

Language-agnostic · EN→RU via M2M100 (offline).

Analysis Pipeline

①Input ▸ ② Whisper ASR ▸ ③spaCy 3.7 ▸ ④ TAM Engine ▾ ⑤ Phrase Builder ◂ ⑥ML Layer ◂ ⑦Validator ◂ ⑧RLE JSON

PWA — Android-installable, fully client-side, no cloud.

Research Journey: What We Tried

CEFR CLASSIFICATION

~~DeBERTa~~ ~~underperformed~~ → ~~BERT~~ ~~same~~ →

XGBoost ✓ 78.3%

ASR

Whisper ✓ WER 8.2%

NOTE GENERATION

~~GPT-4o~~ ~~cost/cloud~~ → ~~T5 raw~~ ~~no generalise~~ →

T5+Templates ✓ ROUGE-L 0.81

TRANSLATION

~~GPT-4o~~ ~~opt-in~~ → **M2M100 ✓** offline

Context–Pattern Interaction

> **"In open systems, context forms patterns.**
> **In closed systems, patterns form context."**

— Discovered empirically during T5 fine-tuning

T5 on raw notation **failed**. Structural templates — **lexically invariant** — were the breakthrough:

❌ `"gave him" → [VP gave][NP him]`
✅ `[V-PAST][IOBJ] → [TAM:past-simple] [VOICE:active]`

The Duality Loop Diagram

🌊 Open System
Training Phase
RLE trees carve the encoder's weight space — the "River Bed" pattern.
Context → Pattern

Open → **T5** → Closed

🔭 Closed System
Inference Phase
Frozen weights refract novel input into CEFR, TAM, pedagogical note.
Pattern → Context

Training: RLE trees → encoder weights

Inference: frozen weights → annotations

Training Data & Corpus Scale

OANC	**30M+ tokens**
UD GUM	**5.2M tokens**
Gutenberg	**1M+ tokens**
MASC	**~500k tokens**

27k→685 unique · **6 CEFR** A1→C2

Live Demo

www.el-a.uk

Try it in your browser

Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	$$$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	**High**	**Free**	✓
XGBoost CEFR	78.3%	Free	✓

System Validation

- **O1:** spaCy + TAM → schema-valid RLE ✓
- **O2:** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- **O3:** Visualizer + JSON/CSV export ✓
- **O4:** Android PWA, fully client-side ✓

Performance Metrics

78.3%	0.81	8.2%
CEFR Accuracy	**ROUGE-L**	**ASR WER**
XGBoost ensemble	T5 fine-tuned	Whisper base

Conclusions

- AI annotation makes authentic English **accessible without simplification**.
- The **Context↔Pattern duality** generalises to any open→closed system; T5 placeholders are **lexically invariant**.

Future Directions

- **Tauri 2 Desktop App** — native .deb/Windows with bundled ML models.
- **M2M100 LoRA Fine-Tuning** — domain-specific EN→RU refinement.
- **Translation Quality Gates** — ChrF scorer, M2M100 review queue.
- **Real-time Streaming ASR** — Whisper streaming for live practice.

References

Acknowledgements

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